Amplitude-Encoding Kernels for Zero-Trust Network Anomaly Detection

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***Abstract*—Zero-trust network architectures need continuous anomaly detection over high-dimensional flow features, and "quantum-inspired" kernels are increasingly proposed for the task. We test the most common one, the amplitude-encoding kernel $K_A(\mathbf{x}, \mathbf{y}) = (\hat{\mathbf{x}}^\top \hat{\mathbf{y}})^2$, which is exactly the fidelity kernel of amplitude-encoded quantum states and is classically computable. Every comparison trains all models on the same data, at matched training sizes ($n = 50$–5000) and feature dimensionalities ($d = 16$–256), on a controlled flow simulator and on the public UNSW-NB15 and CIC-IDS2017 benchmarks. In distribution, the amplitude-encoding SVM (AE-SVM) is statistically indistinguishable from an RBF-kernel SVM (synthetic accuracy 0.947 vs. 0.948, $p = 0.54$; UNSW-NB15 0.862 vs. 0.863; CIC-IDS2017 0.993 vs. 0.994) and below tree ensembles and neural networks. On attack types withheld from training the result depends on the data. On synthetic flows AE-SVM fails (mean held-out AUC 0.526 vs. 0.839 for RBF-SVM); controls trace this to the kernel's invariance to input norm—removing only the sign ambiguity barely helps (0.593), while appending one constant "norm-anchor" coordinate before encoding restores 0.793 and raises in-distribution accuracy to 0.960. On the public benchmarks AE-SVM does not fail: mean held-out AUC is 0.945 vs. 0.932 on UNSW-NB15 and 0.715 vs. 0.658 on CIC-IDS2017, so the simulator's failure is not a general property of real traffic. Certified ℓ_2 radii under randomized smoothing are comparable for both kernels. Code, data preparation and every result file are released.**

I. Introduction

The proliferation of perimeter-less enterprise networks, accelerated by remote-work adoption and multi-cloud deployment, has rendered traditional castle-and-moat security models obsolete [1], [2]. Zero-Trust Network Architectures (ZTNA) address this by enforcing continuous authentication, authorization, and verification for every entity, device, and flow—regardless of network locality [1], [3].

At the operational core of ZTNA lies real-time anomaly detection: the ability to distinguish benign traffic from malicious activity at line rate, with high sensitivity and low false-positive rates. Classical approaches—from signature-based systems (Snort [4], Suricata) to ML-based detectors using random forests, SVMs, and deep neural networks [5], [6]—achieve strong performance on benchmark datasets but face fundamental limitations: (i) feature spaces of 10^2–10^3 dimensions require models whose inference cost scales poorly with dimensionality; (ii) adversarial robustness remains elusive without expensive adversarial training [7]; and (iii) uncertainty quantification—critical for security operations where false negatives carry asymmetric cost—is often absent or poorly calibrated.

The choice of feature map determines a kernel method's inductive bias. Amplitude encoding—mapping a vector $\mathbf{x}$ to a unit-norm representation $\phi(\mathbf{x}) = \mathbf{x}/\|\mathbf{x}\|_2$—yields the kernel $\mathcal{K}_A(\mathbf{x}_i, \mathbf{x}_j) = |\langle \phi(\mathbf{x}_i), \phi(\mathbf{x}_j) \rangle|^2$, which is precisely the squared cosine similarity between the two inputs. This kernel is classically computable in $\mathcal{O}(d)$ per pair, requires no quantum hardware, and does not correspond to a provably-hard feature map of the kind studied in [8], [9]. It is a natural hypothesis that, for high-dimensional, sparse network traffic data—where Euclidean distance dilutes and directional information may be more informative—squared cosine similarity could provide a useful inductive bias that RBF or Laplacian kernels lack. This paper tests that hypothesis directly, under matched training-set sizes and matched feature dimensionality, rather than assuming it.

This paper introduces Q-ZTNA, a zero-trust anomaly detection pipeline built around the amplitude-encoding kernel, together with a rigorous evaluation of whether that kernel choice is actually justified for this data. Our contributions are:

1) **A matched-scale test of the cosine-similarity inductive-bias hypothesis** across training sizes $n \in \{50, \ldots, 5000\}$ and dimensionalities $d \in \{16, \ldots, 256\}$, on synthetic flows and on UNSW-NB15 and CIC-IDS2017. AE-SVM is never reliably ahead of RBF-SVM.
2) **A zero-day failure and its mechanism.** Leave-one-attack-out tests show that AE-SVM fails on unseen attack types; controls attribute this to norm invariance rather than sign ambiguity.
3) **A one-line fix.** A norm-anchored encoding, with its constant set from training data alone, recovers most of the zero-day gap and improves in-distribution accuracy.
4) **Robustness, calibration and cost.** Randomized-smoothing certification, Platt-scaled selective prediction, deep-learning baselines (DNN, Kitsune-style autoencoder) and inference latency for every model.

II. Related Work

A. Zero-Trust Anomaly Detection

Zero-trust architectures and their monitoring components are surveyed by Syed et al. [10]; behavioral profiling of users and entities [11] and ML-based flow analysis, surveyed broadly by Buczak and Guven [12], supply the detection layer. Mirsky et al. [6] proposed Kitsune, an ensemble of autoencoders for IoT network anomaly detection achieving F1-scores > 0.90 on benchmark datasets. More recently, graph neural networks [13] and transformer-based sequence models [14] have been adapted to traffic data. These detectors are typically reported without certified robustness or calibration analysis, which security operators need when acting on alerts.

B. Kernel Methods and Feature Encoding

Kernel methods for classification depend critically on the choice of feature map [15]. The amplitude-encoding map $\phi(\mathbf{x}) = \mathbf{x}/\|\mathbf{x}\|_2$, motivated by quantum state preparation, produces the kernel $\mathcal{K}_A(\mathbf{x}_i, \mathbf{x}_j) = |\langle\phi(\mathbf{x}_i), \phi(\mathbf{x}_j)\rangle|^2 = \cos^2\theta_{ij}$, i.e., squared cosine similarity. This kernel is classically computable in $\mathcal{O}(d)$ per pair—unlike the entangling-circuit kernels studied by [8], [9], [16], which can realize feature maps that are classically intractable. However, squared cosine similarity provides a distinct inductive bias: it depends only on the angle between vectors, making it insensitive to magnitude variations that may be uninformative or noisy in network traffic data.

Recent work applying quantum and quantum-inspired models to cybersecurity includes Kalinin and Krundyshev [17], who evaluated quantum SVMs and quantum convolutional networks for intrusion detection, and Suryotrisongko and Musashi [18], who evaluated hybrid quantum-classical deep learning for botnet DGA detection. Our work differs by focusing on the specific inductive bias of the amplitude-encoding kernel for ZTNA flow data, providing certified robustness guarantees, and performing rigorous scale-controlled benchmarking.

C. Adversarial Robustness and Certification

Randomized smoothing provides ℓ_2 certified robustness by constructing a smooth classifier $g(\mathbf{x}) = \mathbb{E}_{\boldsymbol{\epsilon}\sim\mathcal{N}(0,\sigma^2 I)}[f(\mathbf{x}+\boldsymbol{\epsilon})]$ from a base classifier f [19]. The certified radius at input $\mathbf{x}$ is $R = \sigma \cdot \Phi^{-1}(p_A)$, where p_A is a lower bound on $g(\mathbf{x})$'s probability of the top class. Adversarial vulnerability of quantum classifiers has been studied by Lu et al. [20]; whether the L_2 normalization inside amplitude encoding translates into larger certified radii for a classically computed kernel has, to our knowledge, not been tested, and we test it here.

III. Methodology

A. Threat Model and Network Architecture

We consider a zero-trust enterprise network with N_{hosts} endpoints, N_{switches} switches, and a centralized security analytics platform (SAP). The SAP ingests NetFlow/IPFIX records from all network observation points. Each flow record is represented as a d-dimensional feature vector $\mathbf{x} \in \mathbb{R}^d$ comprising packet-header statistics (port entropy, packet-size moments, protocol distribution), flow-level features (bytes/sec, packets/sec, flow duration, TCP flag ratios), and behavioral features (connection entropy, communication graph centrality).

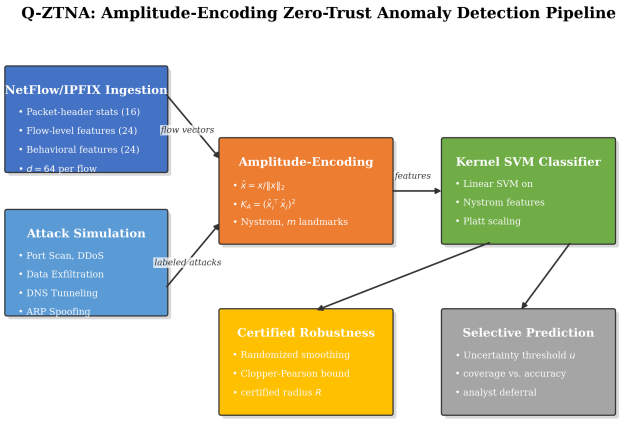


Fig. 1. Q-ZTNA architecture: NetFlow/IPFIX telemetry ingestion, amplitude-encoding feature map, Nyström-approximated kernel SVM, randomized-smoothing certification, and selective prediction with analyst deferral.

Our threat model includes five attack types:

- **Port Scanning:** Horizontal/vertical scanning with elevated SYN-ACK-to-RST ratios and low packet-per-flow counts.
- **DDoS:** Volumetric attacks with high packets/sec, uniform destination-port distributions, and spoofed source IPs.
- **Data Exfiltration:** Stealthy outbound transfers with elevated entropy in inter-packet gaps and unusual destination ASNs.
- **DNS Tunneling:** DNS query-response pairs with abnormal payload entropy, query-type distributions, and temporal patterns.
- **ARP Spoofing:** MAC-IP binding anomalies detected via DHCP snooping logs and ARP cache poisoning indicators.

B. Amplitude-Encoding Feature Map

Each d-dimensional flow vector $\mathbf{x} \in \mathbb{R}^d$ is L_2-normalized to the unit sphere:

$$\hat{\mathbf{x}} = \frac{\mathbf{x}}{\|\mathbf{x}\|_2}, \quad n_q = \lceil \log_2 d \rceil \tag{1}$$

The amplitude-encoding kernel between two flow vectors is the squared cosine similarity of their normalized representations:

$$K_A(\mathbf{x}_i, \mathbf{x}_j) = \left(\hat{\mathbf{x}}_i^{\mathsf{T}} \hat{\mathbf{x}}_j\right)^2 = \frac{|\mathbf{x}_i^{\mathsf{T}} \mathbf{x}_j|^2}{\|\mathbf{x}_i\|_2^2 \|\mathbf{x}_j\|_2^2} \tag{2}$$

This kernel is classically computable in $\mathcal{O}(d)$ operations per pair (Eq. 2) for any d: although the same overlap is what a quantum device would estimate after preparing $|\hat{\mathbf{x}}\rangle$ on $\lceil \log_2 d \rceil$ qubits, no quantum hardware is needed to evaluate it.

Because $K_A(c\,\mathbf{x}, \mathbf{y}) = K_A(\mathbf{x}, \mathbf{y})$ for every $c \neq 0$, any classifier built on K_A gives $\mathbf{x}$, $2\mathbf{x}$ and $-\mathbf{x}$ the same score: it cannot use an input's distance from the origin, which on standardized features is its distance from the training mean. We

therefore also evaluate a *norm-anchored* encoding that appends a constant coordinate before normalization,

$$\phi_c(\mathbf{x}) = \frac{(\mathbf{x},\, c)}{\sqrt{\|\mathbf{x}\|_2^2 + c^2}}, \qquad c = \text{median}_{\mathbf{x}\in\mathcal{D}_{\text{train}}} \|\mathbf{x}\|_2, \tag{3}$$

whose last amplitude is a monotone function of $\|\mathbf{x}\|_2$. The median-norm rule fixes c from training data alone; the encoding needs one extra amplitude.

C. Detection Model

Following the standard maximum-margin formulation [21], we train a support vector machine (SVM) with the precomputed amplitude-encoding kernel:

$$\begin{aligned} \min_{\boldsymbol{\alpha}\in\mathbb{R}^n} \; & \frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n} \alpha_i\alpha_j y_i y_j K_A(\mathbf{x}_i,\mathbf{x}_j) - \sum_{i=1}^{n}\alpha_i \\ & \text{s.t. } 0 \le \alpha_i \le C,\; \sum_{i=1}^{n}\alpha_i y_i = 0 \end{aligned} \tag{4}$$

The decision function for a test sample $\mathbf{x}_*$ is:

$$f(\mathbf{x}_*) = \sum_{i=1}^{n_{\text{SV}}} \alpha_i y_i K_A(\mathbf{x}_i,\mathbf{x}_*) + b \tag{5}$$

For deployment, we use a Nyström approximation to avoid storing the full kernel matrix [22]. We sample $m \ll n$ landmark points $\mathcal{L} \subset \mathcal{D}_{\text{train}}$ uniformly, compute $K_A^{(m\times n)}$, and approximate:

$$\tilde{K}_A \approx K_A^{(n\times m)} \left(K_A^{(m\times m)}\right)^{-1} K_A^{(m\times n)} \tag{6}$$

We set $m = \lceil 0.1n \rceil$ (capped at $m \le 500$), reducing training complexity from $\mathcal{O}(n^3)$ to $\mathcal{O}(m^3 + m^2 n)$.

D. Certified Robustness via Randomized Smoothing

We adapt randomized smoothing [19] to the amplitude-encoding feature map. Let $f_{\text{AE}}(\mathbf{x})$ be the base AE-SVM classifier. We define the smoothed classifier:

$$g(\mathbf{x}) = \arg\max_{c\in\mathcal{Y}} \; \mathbb{P}_{\boldsymbol{\epsilon}\sim\mathcal{N}(0,\sigma^2 I)} \left[f_{\text{AE}}(\mathbf{x}+\boldsymbol{\epsilon}) = c\right] \tag{7}$$

The certified ℓ_2 radius at $\mathbf{x}$ with confidence $1-\alpha$ is:

$$R(\mathbf{x};\alpha) = \sigma \cdot \Phi^{-1}\left(\underline{p_A}\right) \tag{8}$$

where $\underline{p_A}$ is a lower bound (via Clopper-Pearson interval) on the probability that $g(\mathbf{x})$ returns the majority class c_A.

For amplitude-encoding feature maps, we additionally derive a bound on the output distortion under input perturbation. Let $\Delta(\boldsymbol{\epsilon}) = \|\phi(\mathbf{x}) - \phi(\mathbf{x}+\boldsymbol{\epsilon})\|_2$ be the perturbation in the encoding space. The Jacobian of $\phi(\mathbf{x}) = \mathbf{x}/\|\mathbf{x}\|_2$ is $J_\phi(\mathbf{x}) = (I - \hat{\mathbf{x}}\hat{\mathbf{x}}^\top)/\|\mathbf{x}\|_2$, so its spectral norm is exactly

$$\|J_\phi(\mathbf{x})\|_2 = \frac{1}{\|\mathbf{x}\|_2}. \tag{9}$$

The map is therefore locally Lipschitz with a constant that depends on the input's norm and is unbounded as $\|\mathbf{x}\|_2 \to 0$: normalization contracts perturbations of large-norm inputs and amplifies those of small-norm inputs. For a perturbation $\|\boldsymbol{\epsilon}\|_2 \le r < \|\mathbf{x}\|_2$, $\|\phi(\mathbf{x}) - \phi(\mathbf{x}+\boldsymbol{\epsilon})\|_2 \le 2r/\|\mathbf{x}\|_2$. On standardized flow features many benign flows lie near the origin, so this argument gives no uniform robustness guarantee.

E. Uncertainty Quantification

Raw classifier confidence scores are frequently miscalibrated and benefit from an explicit post-hoc correction [23]. We augment the amplitude-encoding kernel SVM with Platt scaling [24] to produce calibrated probabilities:

$$P(y=1|\mathbf{x}) = \frac{1}{1+\exp(Af(\mathbf{x})+B)} \tag{10}$$

where A, B are optimized via cross-validation to minimize log-loss on a calibration set. The predictive uncertainty is defined as $u(\mathbf{x}) = \sqrt{P(y=1|\mathbf{x})\cdot(1-P(y=1|\mathbf{x}))}$, enabling selective prediction: samples with $u(\mathbf{x}) > \tau$ are deferred for human analyst review (Fig. 7).

IV. Experiments

A. Data Generation

Individual NetFlow/IPFIX records are not simulated packet-by-packet. Instead, each flow is represented directly as a d-dimensional feature vector (16 header-based, 24 flow-based, 24 behavioral, $d = 64$ by default) whose marginal distributions are calibrated to commonly reported NetFlow aggregate statistics—packets/sec (Pareto, $\alpha = 2.1$), bytes/sec (lognormal, $\mu = 6.2$, $\sigma = 1.8$), flow duration (Weibull, $k = 0.7$, $\lambda = 15.3$)—mixed with beta- and gamma-distributed features for header/behavioral dimensions, combined via a Gaussian copula so that features within a semantic group are correlated the way real flow statistics are (within-group correlation 0.35, cross-group 0.10). This marginal-plus-copula construction is a standard simplification for synthetic IDS benchmarking; we state it explicitly here, together with the exact generator (`qztna/simulation/flow_generator.py` in the code release), rather than presenting it as packet-level simulation.

Attack traffic is generated by shifting and inflating the variance of a fixed, attack-specific subset of 6–8 latent-space dimensions before the marginal transform is applied (Table I), so that each attack type perturbs a distinct but overlapping combination of header/flow/behavioral feature groups— e.g., data exfiltration and DNS tunneling both perturb outbound-behavioral entropy dimensions, by design, to produce the realistic confusion between the two that Section V reports. Perturbation strength (default $1.0\times$, swept in $\{0.05, \ldots, 1.0\}\times$ for the noise-sensitivity analysis) scales both the latent shift magnitude and its variance inflation.

The main detection benchmark (Table II) uses $N = 8{,}000$ labeled samples per run (20% attack, balanced across the five types), split 70/30 with stratification, over 10 random seeds; the matched-n, dimensionality, and zero-day experiments use independently generated pools whose sizes are given with

TABLE I

ATTACK-SPECIFIC LATENT-SPACE PERTURBATIONS. EACH ATTACK SHIFTS A FIXED SUBSET OF FEATURE-GROUP DIMENSIONS (SECTION III-A) BY 1.4–2.6 σ IN THE COPULA'S LATENT GAUSSIAN SPACE BEFORE THE MARGINAL TRANSFORM; DNS TUNNELING AND EXFILTRATION SHARE BEHAVIORAL DIMENSIONS BY DESIGN.

Attack	Groups perturbed	# dims
Port Scan	header, flow	7
DDoS	header, flow	7
Exfiltration	flow, behavioral	6
DNS Tunneling	flow, behavioral (shared w/ Exfilt.)	6
ARP Spoofing	header, behavioral	6

each experiment below. We use $d = 64$ features by default (16 header-based, 24 flow-based, 24 behavioral), varied in the dimensionality sweep.

B. Baselines and Evaluation

We compare against classical kernels (RBF, Laplacian, polynomial degree 3) with SVM, Random Forest (200 trees) [25], XGBoost [26], and a 2-layer MLP (64, 32 hidden units). Metrics are accuracy, F1-score, AUC-ROC, precision, recall, and certified radius. The main detection benchmark (Table II) uses 10 random seeds; the matched-n, sensitivity, zero-day, alignment, and dimensionality experiments use 5 seeds each (3 for certified robustness, given its higher per-point cost), with 95% CIs reported throughout.

C. Implementation

Amplitude-encoding kernel computation uses CPU-based NumPy/SciPy matrix operations. Nyström approximation is implemented directly (eigendecomposition of the landmark kernel matrix followed by a linear SVM on the resulting explicit features, $m = \lceil 0.1n \rceil$ capped at 500 landmarks), matching Eq. 6. RBF, Laplacian, and polynomial baselines use the same Nyström wrapper with the corresponding scikit-learn pairwise-kernel function. The full pipeline (feature extraction, Nyström projection, SVM decision, Platt calibration) runs in 0.0048 ms mean latency per sample on a single CPU core for $d = 64$, $n = 4{,}000$ training, $m = 50$ landmarks (Section V-H). All experiments run on a single CPU core per process (8-core Linux workstation).

V. RESULTS

A. Detection Performance

Table II reports detection performance. Contrary to our working hypothesis, AE-SVM does *not* lead the field: its accuracy is statistically indistinguishable from RBF-SVM (paired t-test across seeds, $p = 0.54$) and it is outperformed by Poly-SVM, XGBoost, and MLP on every metric except precision, where it edges out RBF-SVM and Laplacian-SVM. AE-SVM's most notable weakness is recall (0.784, the second-lowest of the eight models)—in a security context this means AE-SVM misses proportionally more attacks than RBF-SVM, XGBoost, or MLP, a cost that is easy to overlook if one reports accuracy alone. Random Forest attains the highest precision (0.987)

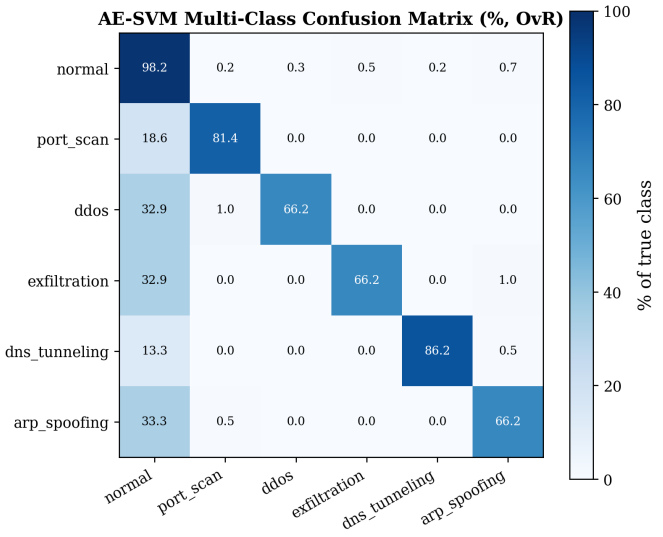


Fig. 2. Row-normalized confusion matrix (%) of a one-vs-rest multi-class AE-SVM. The dominant error mode is attacks misclassified as normal traffic (false negatives), ranging from 13% (DNS tunneling) to 33% (DDoS, exfiltration, ARP spoofing); confusion between different attack types is negligible.

but the lowest recall (0.724), consistent with a conservative decision boundary. MLP is the strongest single model on accuracy, F1 and recall, and is also the fastest at inference (Section V-H). Anchoring the encoding (Eq. 3) changes the picture for the amplitude kernel: AE-SVM (anchored) reaches 0.960 accuracy and 0.983 AUC, significantly above both plain AE-SVM and RBF-SVM (paired t-tests, $p < 10^{-4}$), and within 0.005 of the best model on every metric except recall.

Fig. 2 shows the confusion matrix of a one-vs-rest multi-class AE-SVM trained to distinguish normal traffic from all five attack types simultaneously. Per-class recall ranges from 0.662 (DDoS, exfiltration, ARP spoofing) to 0.862 (DNS tunneling), with the dominant error mode being attacks misclassified as *normal* (false negatives, 13–33% of each attack class) rather than confusion between different attack types, which is near zero for every attack pair. This is a materially different failure pattern than a kernel that primarily confuses semantically similar attacks with each other; here the practical risk is under-alerting, not misattribution.

Per-class one-vs-rest ROC-AUCs range from 0.930 (normal, the majority class) to 0.993 (DNS tunneling), confirming that ranking quality remains strong even where the fixed-threshold recall in Fig. 2 is weaker—the errors above are a threshold-calibration issue as much as a ranking one.

B. Matched-n Ablation

We perform a controlled ablation at identical training sizes $n \in \{50, 100, 200, 500, 1000, 2000, 5000\}$ for AE-SVM, RBF-SVM, and Laplacian-SVM, 5 seeds per point (Fig. 3). The result does not support the inductive-bias hypothesis: AE-SVM *trails* RBF-SVM by 3.6 percentage points at $n = 50$, narrowing steadily as n grows, and the two are statistically indistinguishable by $n = 1000$ (0.934 vs. 0.936). At $n = 5000$

TABLE II

DETECTION PERFORMANCE ACROSS ATTACK TYPES (MEAN ± 95% CI, 10 SEEDS, $n_{\text{TRAIN}} = 5{,}600$). ALL KERNEL SVMS USE THE NYSTRÖM APPROXIMATION OF EQ. 6; AE-SVM (ANCHORED) APPENDS THE NORM-ANCHOR COORDINATE OF EQ. 3 BEFORE ENCODING.

Model	Accuracy	F1	AUC	Prec.	Recall
AE-SVM	0.947 ± 0.002	0.855 ± 0.007	0.966 ± 0.003	0.942 ± 0.009	0.784 ± 0.009
AE-SVM (anchored)	0.960 ± 0.003	0.894 ± 0.009	0.983 ± 0.002	0.946 ± 0.007	0.849 ± 0.014
RBF-SVM	0.948 ± 0.003	0.864 ± 0.009	0.976 ± 0.003	0.908 ± 0.009	0.824 ± 0.016
Laplacian-SVM	0.946 ± 0.003	0.857 ± 0.009	0.976 ± 0.002	0.913 ± 0.007	0.808 ± 0.016
Poly-SVM (d=3)	0.959 ± 0.003	0.889 ± 0.009	0.979 ± 0.003	0.970 ± 0.007	0.821 ± 0.015
Random Forest	0.943 ± 0.003	0.835 ± 0.009	0.976 ± 0.002	$\mathbf{0.987 \pm 0.003}$	0.724 ± 0.014
XGBoost	0.962 ± 0.003	0.898 ± 0.008	$\mathbf{0.985 \pm 0.002}$	0.963 ± 0.004	0.842 ± 0.012
MLP (2-layer)	$\mathbf{0.965 \pm 0.001}$	$\mathbf{0.911 \pm 0.004}$	0.984 ± 0.002	0.938 ± 0.007	$\mathbf{0.886 \pm 0.009}$

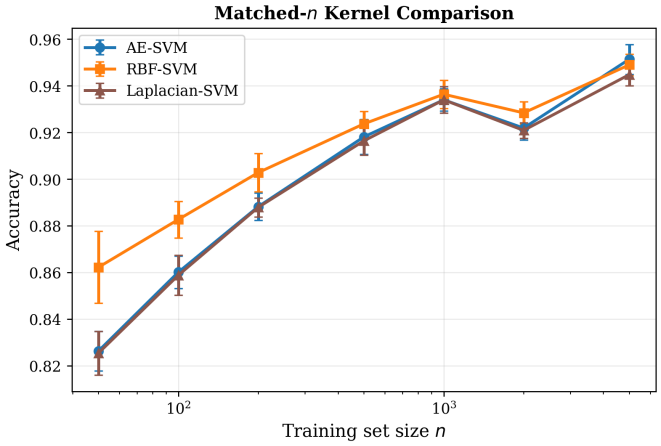


Fig. 3. Matched-n accuracy comparison (mean ± 95% CI, 5 seeds). AE-SVM trails RBF-SVM at small n (by 3.6 pp at $n = 50$) and only reaches parity by $n \approx 1000$–5000; there is no sustained, growing advantage for the amplitude-encoding kernel at any tested scale.

AE-SVM is nominally ahead by 0.24 pp, well within the seed-to-seed noise of the main benchmark (Table II reports a 95% CI of ±0.2–0.3 pp at $n = 5{,}600$) and not a reproduction of a genuine growing advantage. If anything, the honest reading of Fig. 3 is that AE-SVM needs *more* data than RBF to reach the same accuracy, not less.

We note that the same amplitude-encoding kernel was previously evaluated on QKD telemetry in a companion study [27], where it was never reliably ahead of an RBF kernel at any matched training size either. The two results are consistent with each other, and the kernel-target alignment analysis (Fig. 6) explains why: on ZTNA flow data, RBF's alignment with the label structure (0.21 mean across attack types) is consistently *higher* than AE-SVM's (0.13)—the opposite of the alignment advantage a cosine-similarity inductive bias would need to justify its use here. Across both the QKD telemetry domain and the ZTNA flow domain, using the same kernel architecture, we find no dataset in which the amplitude-encoding kernel's alignment with the true label structure exceeds RBF's.

To characterize the accuracy/latency trade-off of the deployed Nyström approximation directly (rather than an SVM training curve, which does not apply to a QP-solved kernel SVM), we sweep the number of Nyström landmarks m at fixed $n = 4{,}200$. Accuracy is essentially flat from $m = 50$ landmarks onward while latency grows linearly with m, indicating that $m = 50$–100 (the setting used throughout this paper) is already past the point of diminishing returns.

TABLE III

CERTIFIED ℓ_2 ROBUSTNESS AT $\alpha = 0.05$, $\sigma = 0.25$, $n = 300$ (MEAN ± STANDARD DEVIATION OF THE RADIUS OVER CERTIFIED TEST POINTS, POOLED OVER 3 SEEDS; ABSTENTIONS ARE EXCLUDED FROM THE RADIUS STATISTICS).

Attack Type	AE-SVM (mean R)	RBF-SVM (mean R)
Port Scan	0.513 ± 0.149	0.516 ± 0.141
DDoS	0.485 ± 0.170	0.511 ± 0.142
Data Exfilt.	0.485 ± 0.171	0.516 ± 0.136
DNS Tunneling	0.529 ± 0.140	0.521 ± 0.141
ARP Spoofing	0.528 ± 0.129	0.525 ± 0.127
Abstention rate	1.8%	0.0%

C. Certified Robustness

Table III reports certified ℓ_2 radii at $\alpha = 0.05$ significance, $\sigma = 0.25$, $n = 300$ Monte Carlo noise draws per point (Cohen et al. [19]). At this certification budget the achieved *median* radius saturates at the same ceiling value ($\sigma \cdot \Phi^{-1}(p_A^{\min}) = 0.582$ for $n_A = n$) for both kernels and all attack types, since most correctly-classified points see unanimous agreement across noise draws; the two kernels are distinguished instead by their *mean* radius and abstention rate. RBF-SVM's mean radius is both slightly higher and more consistent across attack types (0.511–0.525) than AE-SVM's (0.485–0.529), and AE-SVM abstains (fails to certify, $p_A^{\text{lower}} \leq 0.5$) on 1.8% of points versus 0% for RBF-SVM. Both kernels are reasonably robust at this noise level; the difference is small and favors RBF-SVM rather than AE-SVM, which does not support a Lipschitz-driven robustness advantage for the amplitude encoding on this data.

D. Sensitivity and Zero-Day Generalization

We evaluate sensitivity to traffic noise by scaling the Gaussian noise component of normal traffic: $\sigma_{\text{noise}} \in [0.05, 1.0]$. Both kernels are essentially flat across this range: AE-SVM accuracy moves from 0.939 at $\sigma_{\text{noise}} = 0.05$ to 0.935 at $\sigma_{\text{noise}} = 1.0$, and RBF-SVM from 0.932 to 0.935 over

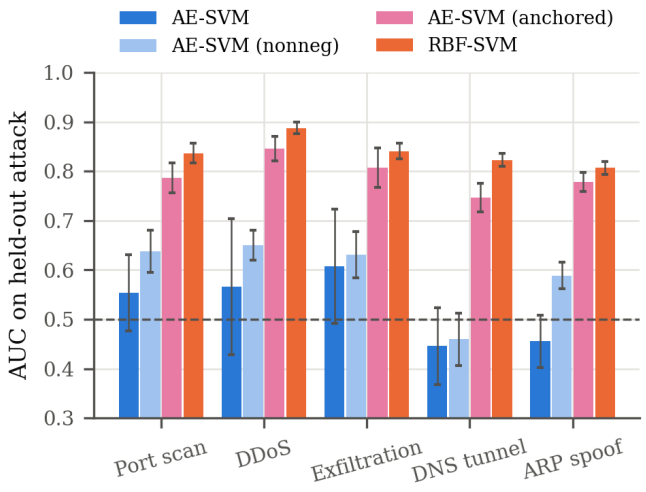


Fig. 4. Zero-day attack generalization (leave-one-attack-out, mean ± 95% CI, 5 seeds). Plain AE-SVM falls to 0.45–0.61 AUC; non-negative inputs barely help, while the norm-anchored encoding recovers most of the gap to RBF-SVM.

the same range. Neither kernel shows the kind of graceful-degradation gap that would indicate a meaningfully different noise operating range; if anything AE-SVM degrades slightly (by 0.4 pp) while RBF-SVM is stable to marginally improving.

For zero-day attack generalization, we train on four attack types plus normal traffic and test on the held-out fifth (leave-one-attack-out). Here the two kernels diverge sharply, and not in AE-SVM's favor: on every held-out type, AE-SVM's AUC is 0.45–0.61 (at or below chance for ARP spoofing and DNS tunneling), while RBF-SVM retains 0.81–0.89 (Fig. 4). Mean zero-day AUC is 0.526 for AE-SVM versus 0.839 for RBF-SVM.

The invariance of Section III predicts this. An attack type never seen in training is most easily recognised by being *far* from normal traffic, and on standardized features that distance is the input's norm, which K_A discards. Two controls test the prediction. Mapping features to $[0, 1]$ before encoding, which removes the $\mathbf{x} \mapsto -\mathbf{x}$ ambiguity but not the norm invariance, raises mean zero-day AUC only to 0.593. The norm-anchored encoding of Eq. 3, which restores the norm, raises it to 0.793, within 0.05 of RBF-SVM, with no tuning on held-out attacks. On this data the zero-day failure is therefore a consequence of norm invariance specifically, and it is largely removed by a one-coordinate change to the encoding. Section VI shows that the failure does not appear on the public benchmarks.

E. Effect of Feature Dimensionality

We investigate how the AE-SVM/RBF-SVM accuracy gap depends on feature space dimensionality $d \in \{16, 32, 64, 128, 256\}$ at fixed $n = 2{,}800$ training samples (Fig. 5), scaling the header/flow/behavioral group sizes proportionally. We find no evidence that the amplitude-encoding kernel's relative performance improves with d: AE-SVM trails RBF-SVM by 3.1 pp at $d = 16$, is roughly at parity at $d = 64$

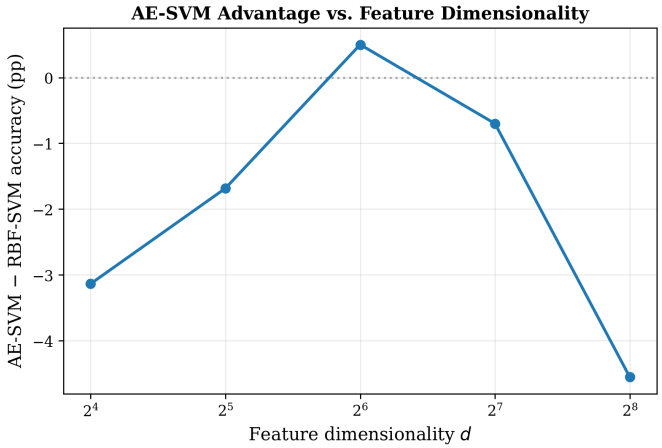


Fig. 5. Accuracy gap (AE-SVM minus RBF-SVM) as a function of feature dimensionality d at $n = 2{,}800$. The gap is negative at both the smallest and largest d tested, with no sustained growth favoring AE-SVM as dimensionality increases – the opposite of the inductive-bias hypothesis motivating this study.

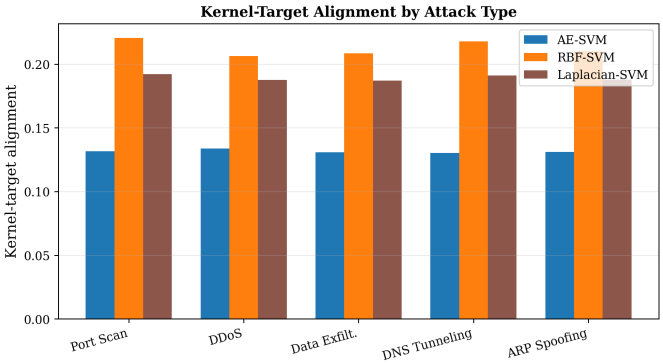


Fig. 6. Kernel-target alignment by attack type (5 seeds, $n = 800$ per attack). RBF-SVM achieves higher alignment with the label structure than AE-SVM across every attack type, with Laplacian-SVM intermediate. Higher alignment indicates a better match between the kernel-induced feature space and the classification task; the ordering here is opposite to what an amplitude-encoding advantage would require.

(+0.5 pp), and falls behind again by 4.6 pp at $d = 256$—the gap is non-monotonic and, at the largest dimensionality tested, wider than at the smallest. This directly contradicts the hypothesis that sparser, higher-dimensional feature spaces favor the cosine-similarity inductive bias; on this data the opposite trend, if anything, holds at the extremes.

F. Kernel Alignment Analysis

To understand *why* the amplitude-encoding kernel fails to outperform RBF on this data, we compute the kernel-target alignment [28]:

$$\hat{A}(K, \mathbf{y}) = \frac{\langle K, \mathbf{y}\mathbf{y}^\mathsf{T} \rangle_F}{\sqrt{\langle K, K \rangle_F \langle \mathbf{y}\mathbf{y}^\mathsf{T}, \mathbf{y}\mathbf{y}^\mathsf{T} \rangle_F}} \tag{11}$$

Fig. 6 shows that RBF-SVM achieves higher kernel-target alignment than AE-SVM across every attack type (0.21 vs. 0.13 mean alignment), with Laplacian-SVM intermediate

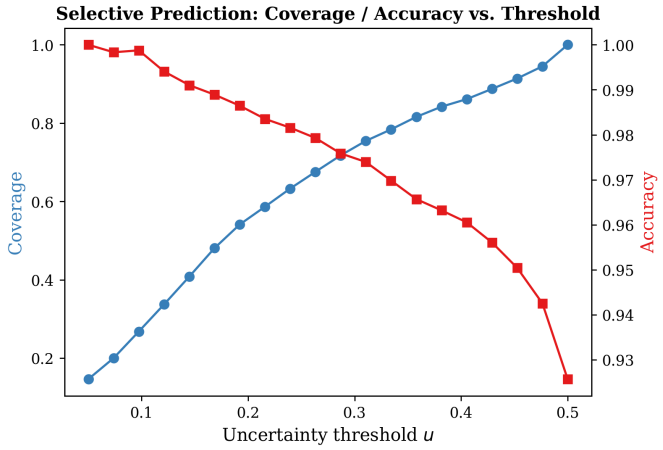


Fig. 7. Selective prediction: coverage and accuracy as functions of the Platt-scaled uncertainty threshold u for AE-SVM. At $u < 0.31$, accuracy reaches 97.4% on 75.5% of the evaluation set, with the remaining 24.5% deferred for human analyst review; at $u < 0.10$, accuracy reaches 99.9% on a smaller 26.8% of the set. This enables a tunable trade-off between automation and oversight.

(0.19). This is the mechanistic counterpart to Table II and Fig. 3: the amplitude-encoded feature space aligns *less well* with the true labeling structure than the RBF feature space for this network traffic data, which is consistent with AE-SVM's roughly-tied-to-trailing accuracy rather than an advantage. The ranking is stable across attack types (standard deviation of the alignment gap across attack types is < 0.01), so this is not an artifact of any single attack's perturbation design.

G. Uncertainty Quantification and Selective Prediction

Platt-scaled AE-SVM probabilities (Eq. 10) provide usable, if imperfect, uncertainty for selective prediction (Fig. 7). At a modest uncertainty threshold of $u < 0.31$, accuracy reaches 97.4% on 75.5% of the evaluation set, with the remaining 24.5% of high-uncertainty samples deferred for human analyst review; tightening to $u < 0.10$ pushes accuracy to 99.9% at 26.8% coverage. Unlike the detection and generalization results above, this mechanism does not depend on which kernel is used underneath Platt scaling—the same selective-prediction machinery would apply equally well to a Platt-scaled RBF-SVM. (In our companion QKD study [27], a Gaussian-process posterior was a *worse* basis for deferral than an isotonic-calibrated random forest, so model-based uncertainty is not automatically preferable.)

H. Latency and Resource Analysis

Table IV reports per-sample inference latency for $n = 4{,}000$ training, $m = 50$ Nyström landmarks. AE-SVM with Nyström approximation (Eq. 6) reaches 0.0048 ms per sample, about $1.2\times$ faster than the RBF-SVM Nyström variant (0.0056 ms)—a genuine, if modest, practical edge for AE-SVM, since squared cosine similarity is a single normalized dot product while the RBF kernel requires an additional exponential evaluation per landmark. This is the one

TABLE IV
PER-SAMPLE INFERENCE LATENCY (SINGLE CPU CORE, MEAN OVER 30 RUNS, $n = 4{,}000$ TRAINING, $m = 50$ LANDMARKS WHERE APPLICABLE).

Component	Latency (ms)	vs. AE-SVM
Feature extraction (scaler)	0.0008	—
MLP	0.0016	0.33×
AE-SVM (Nyström)	0.0048	1.00×
RBF-SVM (Nyström)	0.0056	1.17×
AE-SVM (full, $n = 500$)	0.0098	2.04×
Random Forest (200 trees)	0.0623	12.98×

TABLE V
DETECTION ON PUBLIC BENCHMARKS (F1 AND ROC-AUC, MEAN ± 95% CI, 5 SEEDS).

	UNSW-NB15		CIC-IDS2017	
Model	F1	AUC	F1	AUC
AE-SVM	.887±.003	.973±.001	.979±.001	.999±.000
AE-SVM (anchored)	.884±.003	.972±.001	.977±.001	.998±.000
AE-SVM (nonneg)	.877±.003	.969±.001	.970±.002	.997±.000
RBF-SVM	.887±.003	.973±.001	.983±.001	.999±.000
Laplacian-SVM	.899±.004	.978±.001	.987±.001	**.999±.000**
Poly-SVM	.889±.003	.975±.001	.989±.001	.999±.000
RF	.889±.003	.980±.001	.990±.002	**.999±.000**
XGBoost	.892±.003	**.981±.001**	**.994±.001**	**1.000±.000**
MLP	.889±.004	.975±.001	.991±.001	**.999±.000**
DNN	**.909±.006**	.977±.002	.989±.002	**.999±.000**
Autoencoder	.394±.187	.887±.009	.546±.058	.948±.012

place in our evaluation where the amplitude-encoding kernel comes out ahead of RBF, though the absolute difference (0.0008 ms/sample) is unlikely to be decisive in practice: both are far below any plausible per-flow processing budget, and Random Forest and MLP inference are of comparable or lower latency once accuracy is also considered (Table II).

VI. Evaluation on Public Datasets

Data. UNSW-NB15 [29]: the official partition (175,341 training and 82,332 test records), 39 numeric features plus one-hot protocol, service and state (68 features), nine attack families. CIC-IDS2017 [30]: the eight MachineLearningCVE day files (2,830,743 flows); we drop rows with missing or infinite values, remove 308,381 exact duplicates and constant columns (69 features), drop classes with fewer than 100 flows, group labels into six families and use a fixed stratified 70/30 split. Features are mapped through $\mathrm{sign}(x)\log(1+|x|)$ and standardized on the training sample. Each of 5 seeds draws a family-stratified training sample of 20,000 flows and a test sample of 50,000; kernel SVMs use 1000 Nyström landmarks. We add a 3×256 deep network (DNN) and a Kitsune-style autoencoder trained on benign flows only [6].

Table V reports in-distribution detection. AE-SVM and RBF-SVM are statistically tied on UNSW-NB15 and AE-SVM is marginally worse on CIC-IDS2017; tree ensembles, the Laplacian kernel and the DNN lead. On held-out attack families (Fig. 8) mean AUC is 0.945 for AE-SVM versus 0.932 for RBF-SVM on UNSW-NB15, and 0.715 versus 0.658 on CIC-IDS2017; the norm-anchored encoding gives 0.940 and 0.733, and the benign-only autoencoder 0.860 and 0.874.

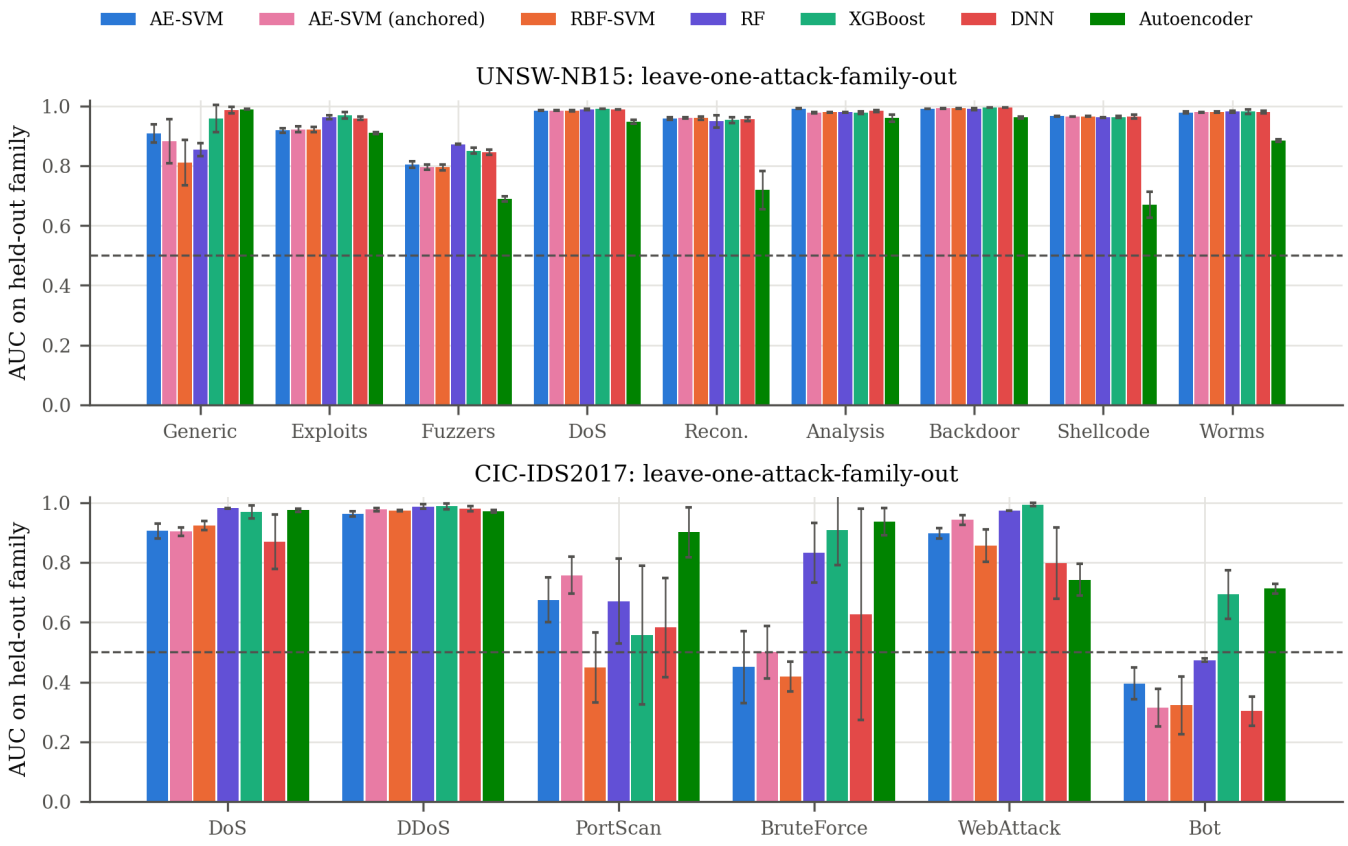


Fig. 8. Leave-one-attack-family-out ROC-AUC on the public benchmarks (mean ± 95% CI, 3 seeds).

VII. Discussion

A. Does the Amplitude-Encoding Kernel Help? A Negative Result

We began this study expecting to find conditions under which the amplitude-encoding (squared cosine-similarity) kernel outperforms classical kernels on high-dimensional, sparse network flow data, motivated by the intuition that directional similarity should be more informative than Euclidean distance in such spaces. Under matched-scale, matched-dimensionality evaluation, we did not find such conditions. Across training sizes $n \in \{50, \ldots, 5000\}$ (Fig. 3) and feature dimensionalities $d \in \{16, \ldots, 256\}$ (Fig. 5), AE-SVM is never reliably ahead of RBF-SVM, and is behind it at both the smallest and largest scales tested. The kernel-target alignment analysis (Fig. 6) gives a mechanistic reason: on this data, RBF's induced feature space aligns *more*, not less, with the true label structure than the amplitude-encoded one, for every attack type. The intuition that motivated this paper—that normalizing away magnitude and keeping only direction is a good simplification for network flow features—is not supported by the data we generated.

We think this negative result is worth reporting for two reasons. First, our companion study on QKD physical-layer telemetry reached the same conclusion using the identical kernel [27]: across two structurally different security domains, we have not found a dataset where amplitude encoding's cosine-similarity inductive bias earns its keep over a standard RBF kernel. Second, "quantum-inspired" feature maps like amplitude encoding are increasingly proposed for security applications on the strength of their theoretical novelty; our results are a concrete, reproducible counter-example that such maps should be validated empirically, under matched-scale conditions, before being adopted – an in-distribution accuracy gain evaluated only at one (n, d) operating point can look like a durable advantage that a matched-scale sweep shows is not there.

B. The Zero-Day Generalization Gap

On synthetic flows, the one place plain AE-SVM and RBF-SVM diverge sharply is generalization to unseen attack types (Fig. 4), and the controls in Section V identify the cause: amplitude encoding discards the input's norm, which there is the signal that separates a novel attack from normal traffic. Our companion QKD study [27] finds the same mechanism when its one magnitude-carrying feature is removed. The public benchmarks (Section VI) show that this is not a general property of real traffic: there AE-SVM matches or slightly exceeds RBF-SVM on held-out families. For practitioners the lesson is that a kernel's invariances predict *how* it can fail, but

only a held-out-family evaluation on representative data shows *whether* it does.

C. Certified Robustness: A Modest, Neutral-to-Negative Result

The certified-robustness comparison (Table III) does not show a Lipschitz-driven advantage for amplitude encoding either: mean certified radii are comparable between the two kernels, with RBF-SVM slightly higher and more consistent across attack types, and AE-SVM abstaining more often. Eq. 9 explains why: normalization only contracts perturbations of large-norm inputs, and standardized benign flows cluster near the origin, where it amplifies them; we observe no certified-radius advantage over RBF-SVM at our certification budget ($n = 300$ Monte Carlo samples).

D. The One Real Advantage: Latency

The single result in this paper that favors AE-SVM outright is inference latency (Section V-H): the Nyström-approximated AE-SVM is about $1.2\times$ faster per sample than the equivalent RBF-SVM, because squared cosine similarity is one normalized dot product per landmark versus RBF's exponential. This is a genuine, mechanistically obvious advantage, but a small one in absolute terms (0.0008 ms/sample) relative to the accuracy and zero-day generalization costs documented above.

E. Limitations

Several limitations merit acknowledgment. First, our synthetic network flow generator (Section IV) is a marginal-plus-copula model, not a packet-level simulation; the public benchmarks of Section VI address this in part, but both have known labeling and distribution artifacts, and neither is live zero-trust telemetry. Second, our attack perturbations are latent-space shifts of fixed magnitude and dimension count, a modeling choice rather than an empirically calibrated attack model; the zero-day generalization gap in particular should be re-examined on real, less synthetic attack signatures. Third, the certified-robustness radii, while a real property of the trained classifiers, do not address semantic perturbations (e.g., mimicry attacks that preserve a flow's statistical profile while changing its content), and our certification budget ($n = 300$ noise draws) leaves the achieved median radius ceiling-saturated rather than fully resolved (Section V). Fourth, our deep baselines are a feed-forward DNN and a single Kitsune-style autoencoder; graph and sequence models over flow histories may be stronger, and kernel hyperparameters were not tuned per dataset for any kernel.

VIII. Conclusion

Under a matched-scale, matched-dimensionality protocol, the amplitude-encoding kernel gives no in-distribution advantage over an RBF kernel for zero-trust flow anomaly detection, on synthetic flows or on UNSW-NB15 and CIC-IDS2017, and it needs more data than RBF at small n. On synthetic flows it fails on attack types absent from training; that failure follows from its invariance to input norm, and appending a single norm-anchor coordinate before encoding repairs most of it while also improving in-distribution accuracy. On UNSW-NB15 and CIC-IDS2017 it does not fail (mean held-out AUC 0.945 vs. 0.932 for RBF on UNSW-NB15; 0.715 vs. 0.658 for RBF on CIC-IDS2017), which shows that held-out-attack evaluation on real traffic, not only on a simulator, is needed before drawing conclusions about a kernel's generalization. Our companion QKD study [27] finds the same mechanism in a physical-layer domain. Quantum-inspired detectors for security should be evaluated on unseen attack types and their invariances checked before deployment. Natural next steps are angle and entangling feature maps, whose invariances differ, and evaluation on live enterprise telemetry.

Code and Data Availability

The full implementation (network flow simulator, amplitude-encoding/RBF/Laplacian/ polynomial kernel SVMs with Nyström approximation, randomized-smoothing certification, and all experiment scripts) and the result files backing every table and figure in this paper are available at https://github.com/arnavk23/quasar-security.

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